

BTECH CSE TY - Foundations of AIML - Mini Project Submission- (2026-27)

Mini Project – Important Instructions

Dear Students,

Please follow the instructions given below for the **Mini Project**:

1. Individual Project:

Each student has to perform the mini project **individually**. Group projects are not allowed.

2. Unique Project Title:

The **project title must be different for every student**. No two students should submit the same project title.

3. Title Submission:

Before finalizing your project, check whether the same or a very similar title has already been submitted by another student.

4. Duplicate Title Rule:

If the **same project title is submitted by more than one student**, the title will be considered for the student whose title was **received/submitted first**. The other student(s) will have to change their project title.

5. First-Come Basis:

Project titles will be considered on a **first-come, first-served basis** according to the date and time of submission.

6. Do Not Copy:

Students should not copy another student's project title, topic, or work. Each student is expected to work independently.

Title Approval:

Students are advised to get their project title approved before starting the implementation.

Important: Once a title has been submitted and accepted, other students cannot claim the same title. So, submit your title as early as possible.

Please take this instruction seriously and ensure that your mini project is completely individual and has a unique title.

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* Indicates required question



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Email *



Record 20240802203@dypiu.ac.in as the email to be included with my response

B.Tech CSE Section Name *



G - Section



H - Section

Student Name *

ANUJ DIXIT

PRN No *

20240802203

Mini Project Title *

SmartSpend: ML-Based Student Expense Pattern and Anomaly Detection System

Problem Staement *

The proposed project aims to analyze student spending data using Machine Learning techniques to identify different spending patterns and detect potentially unusual expense behavior. The system will analyze multiple financial features and group students according to their spending characteristics, helping identify normal and unusual spending patterns.



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Objective of the Project *

The main objective of the project is to develop a Machine Learning-based system that analyzes student expenses, identifies different spending patterns, and detects potentially unusual spending behavior.

Dataset Name *

Student Spending Dataset

Dataset Source/Link *

https://www.kaggle.com/datasets/sumanthnimmagadda/student-spending-dataset?utm_source=chatgpt.com

Number of Records / Rows *

1,000 student records and 17 columns

What type of Machine Learning Problem is your Project? *

Unsupervised Machine Learning – Clustering and Anomaly Analysis

What are the main features and Variables? *

Age, Gender, Year in School, Major, Monthly Income, Financial Aid, Tuition, Housing, Food, Transportation, Books & Supplies, Entertainment, Personal Care, Technology, Health & Wellness, Miscellaneous Expenses, Preferred Payment Method.

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What is the target Output Variable? *

No predefined target variable. The project uses K-Means clustering to identify student spending patterns and assign each student to a spending cluster. Potentially unusual spending behavior will be identified through cluster-based analysis and distance from typical spending patterns.

Which ML Algorithm do you propose? *

- ☐ Logistic Regression
- ☐ Decision Tree
- ☐ KNN
- ☐ SVM
- ☒ K- Means
- ☐ Naive Bayes
- ☐ Random Forest
- ☐ Linear Regression
- ☐ Other:

Expected Result *

The expected result is to identify meaningful groups of students based on their spending behavior using K-Means clustering. The project will determine different spending patterns and identify students whose spending behavior differs significantly from the typical pattern of their cluster.



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Why did you choose this project? *

I chose this project because student financial management is a practical real-world problem. Students spend money across multiple categories, making it difficult to understand their overall spending behavior and identify unusual patterns.

Student Declaration *



I confirmed that this is my individual mini project idea and I have identified a suitable real world dataset for the proposed problem.

Submit

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